Stefano Baldan

Staff Software Engineer · Accessibility & Frontend Platform

Berlin, Germany · EU citizen · stefanobaldan@mail.com
https://stefanobaldan.com · https://github.com/singintime

PROFILE

Staff software engineer with more than ten years of professional experience in frontend development. Five years leading accessibility engineering for Miro's HTML Canvas–based product, from architecture and design through implementation. Drives changes across team boundaries, from shared accessibility packages adopted by over a dozen others to a self-initiated migration through code owned by 26 teams, using AI and LLM tooling to make large-scale work feasible and to help developers write accessible-by-design code. Background in digital signal processing, music, sound synthesis and multimedia computing.

WORK EXPERIENCE

Staff Software Engineer · Miro — Apr 2024 – present

- Removed jQuery from the entire Miro web client: 36 incremental PRs across code owned by 26 teams, deleting 13.8K lines and ending with removal of the runtime global and the dependency. Self-initiated, AI-assisted.
- Consolidated dialog infrastructure to fix recurring focus-management bugs: migrated dialogs across 14 teams' code to the shared accessible dialog package and removed the deprecated imperative dialog service, finding collision guards that had silently never run. Scoped the remaining ~420-consumer migration.
- Created 12 shared packages, including the canvas accessibility layer (adopted by 14 packages), its keyboard-support framework for interactive canvas objects (adopted by 8 packages, from structured documents to AI workflow nodes) and reduced-motion support (13 packages).
- Fixed screen-reader-triggered slowdowns on very large boards: 3x faster bounds computation for 10K-object selections (45 ms to 15 ms), plus bounds caching and debounced announcements.
- Delivered voice dictation end to end: core service, React components, AI chat integration, session token refresh, and screen-reader-verified error handling.
- Led the accessibility DevEx initiative for a monorepo with 400+ active contributors: domain-specific Claude skills distributed through the Claude marketplace, AI-friendly package instructions and specs, and automated accessibility checker rules.
- Steered accessibility engineering towards WCAG 2.2 AA (W3C Recommendation, Oct 2023) and European Accessibility Act compliance (in force since June 2025); documented the architecture in ADRs and specs for keyboard support, virtual menus and multi-step actions.

Frontend Software Engineer · Miro — Nov 2021 – Mar 2024

- Made Miro's HTML Canvas, invisible to standard web accessibility technologies, operable by keyboard, screen reader and voice-control users, driving the work from research and design through implementation.
- Designed and built keyboard navigation across board objects: spatial navigation for free-form, non-grid layouts (nearest object in four directions) and a semantic tab order.
- Developed a DOM overlay that exposes Canvas content to voice-control software, so users can act on board objects by voice.
- Built alternative views of board text and images, ordered by their semantic position on the board.
- Improved rendering performance in the Miro client using the Canvas and WebGL APIs.

Engineering Team Lead · byrd — Jan 2021 – Oct 2021

Logistics company - systems for shipping parcels from origin to destination

- Led and managed a full-stack team of 4 engineers.
- Frontend development in Angular and TypeScript.
- Backend microservices built with Python, Flask, and SQLAlchemy.

Senior Frontend Engineer · byrd — Nov 2019 – Dec 2020

- Developed a single-page application for the company's Warehouse Management System.
- Built reactive pages and UI components in Angular and TypeScript.

Frontend Developer · Contralto Audio — Sep 2017 – Dec 2019

- Developed a 3D editor for the acoustic simulation of loudspeaker positioning in live venues.
- Created custom 3D graphics components and GPU shaders on top of Three.js and WebGL, shipped as a desktop application with Angular and Electron.

Research Assistant · Iuav University of Venice — Jan 2014 – Dec 2017

- Research for the EU project SkAT-VG (Sketching Audio Technologies using Vocalizations and Gestures): https://skatvg.unipa.it/
- Leader of the Work Package on imitation-driven sound synthesis.
- Developed physically-informed sound synthesis and analysis algorithms in Python and C/C++.

Freelance IT Consultant · Self-employed — Oct 2011 – Dec 2013

- Web-based edutainment games for Pearson Italia; interactive media and sound design for digital art installations.

TECHNICAL SKILLS

Frontend: TypeScript, React, JavaScript, HTML Canvas, WebGL, Three.js, design systems, Vitest, Node.js, Electron, Angular

AI: agentic coding with Claude Code, LLM-based tooling for code generation and automated accessibility checks

Accessibility: WCAG 2.2 AA, European Accessibility Act, keyboard and spatial navigation, screen readers, voice-control support, DOM overlays for Canvas content, accessible-by-design practices

DSP & audio: Digital signal processing, physically-informed sound synthesis and analysis, audio and multimedia computing

Backend & other: Python, Flask, SQLAlchemy, C/C++

Engineering practice: Architecture and technical strategy, large-scale monorepo migrations, ADRs, RFCs, technical documentation

EDUCATION

MSc in Economics · University Niccolò Cusano — 2017 – 2020

Main subjects: Environmental Economics, Statistics, Accounting. Final grade: 110/110.

BSc & MSc in Computer Science for New Media Communications · University of Milan — 2005 – 2011

Main subjects: Software Engineering, Digital Signal Processing, Distributed Systems, Human-Computer Interaction, Music Information Retrieval, Semiotics. Final grade: 110/110 cum laude.

RELEVANT PROJECTS

- **Audio plugins** - independent line of audio plugins built on physically and statistically grounded DSP (parametric and analytical methods), distributed via Polar.sh. https://stefanobaldan.com/plugins/
- **vodit** - minimalist dependency injector. No frills, no dependencies, no decorators. https://singintime.github.io/vodit
- **SDT (Sound Design Toolkit)** - open-source C library for physically-based sound synthesis; developer and maintainer, 2014–2017. https://github.com/SkATVG/SDT
- **MaIs** - performance presented in Milan in 2013. Co-author. https://vimeo.com/86330894
- **Sonic Tennis** - audio-only game for iOS devices. https://youtu.be/0uxqhCS0GPg
- **GlobeMX** - interactive HTML5 media player. Master's thesis project. http://emipiu.di.unimi.it
- **IPCapture** - Processing library for acquiring MJPEG live streams. https://github.com/singintime/ipcapture
- **Musica Sull'Acqua** - sonification of the civic aquarium of Milan. https://ludovico.lim.di.unimi.it/download/papers/SMC2012.pdf
- **Moonlanding** - interactive installation, Festival della Creatività, Florence, 2009. Sound designer. https://youtu.be/kEERdTvCbYI

LANGUAGES

Italian (native) · English (C2) · French (C1) · German (B1)

OTHER

Pianist, mandolinist and singer; home-recording engineer. Amateur traditional folk dancer, juggler, and fencer.